Bellabeat Case Study: FitBit Fitness Tracker Data Analysis

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Introduction

This project aims to analyze smart device data to gain insight into how consumers are using the smart devices of *Bellabeat*, a high-tech manufacturer of health-focused products for women. The data is gathered from `https://www.kaggle.com/datasets/arashnic/fitbit/data`.

After cleaning the data using `MySQL` and generating the images using `Tableau`, several categories need to clarify for future discussion:

1. *BMI category*:

   Thinnish : BMI < 18.5 **Normal** : $18.5 \leq$ BMI < 25 **Overweight** : $25 \leq$ BMI < 30 **Obesity** : BMI ≥ 30

2. *User category*:

   HighlyActiveUser : Average Steps ≥ 10000 **ModeratelyActiveUser** : $7500 \leq$ Average Steps < 10000
   GenerallyActiveUser : $5000 \leq$ Average Steps < 7500 **LightlyActiveUser** : Average Steps < 5000

The analysis covers the data from 11 users who both record their activity and body information, and I gathered several key findings for company's future development:

1. **Significant polarization in user activity levels**: Highly active users (User 1, User 9, User 11) has an average over 11,000 daily steps, while lightly active users (User 2, User 5, User 6, User 10) has an average fewer than 5,000 steps. This gap shows extreme differences in exercise habits.

2. **Sedentary behavior dominates users' daily activity**: Most users spend a significant amount of their daily time in a sedentary state. Even highly active users have a sedentary time proportion exceeding 65%. It's crucial to set "reducing sedentary time and increasing exercise time" as the core direction for the product to inform users to maintain a healthier habit.

3. **Strong correlation between exercise intensity and health status**: Users with normal BMI have a significantly higher proportion of highly active exercise than overweight and obese users. It's evident that insufficient exercise is the main cause of weight gain.

4. **Improvement in data quality**: Some users have extreme outliers in their data, and it's necessary to improve device calibration and data validation mechanisms.

User Activity Level Segmentation and Overall Distribution

1. Activity Level Segmentation and Proportion

After analyzing user activity and activity time data, several useful figures are shown below based on the cleaned data:

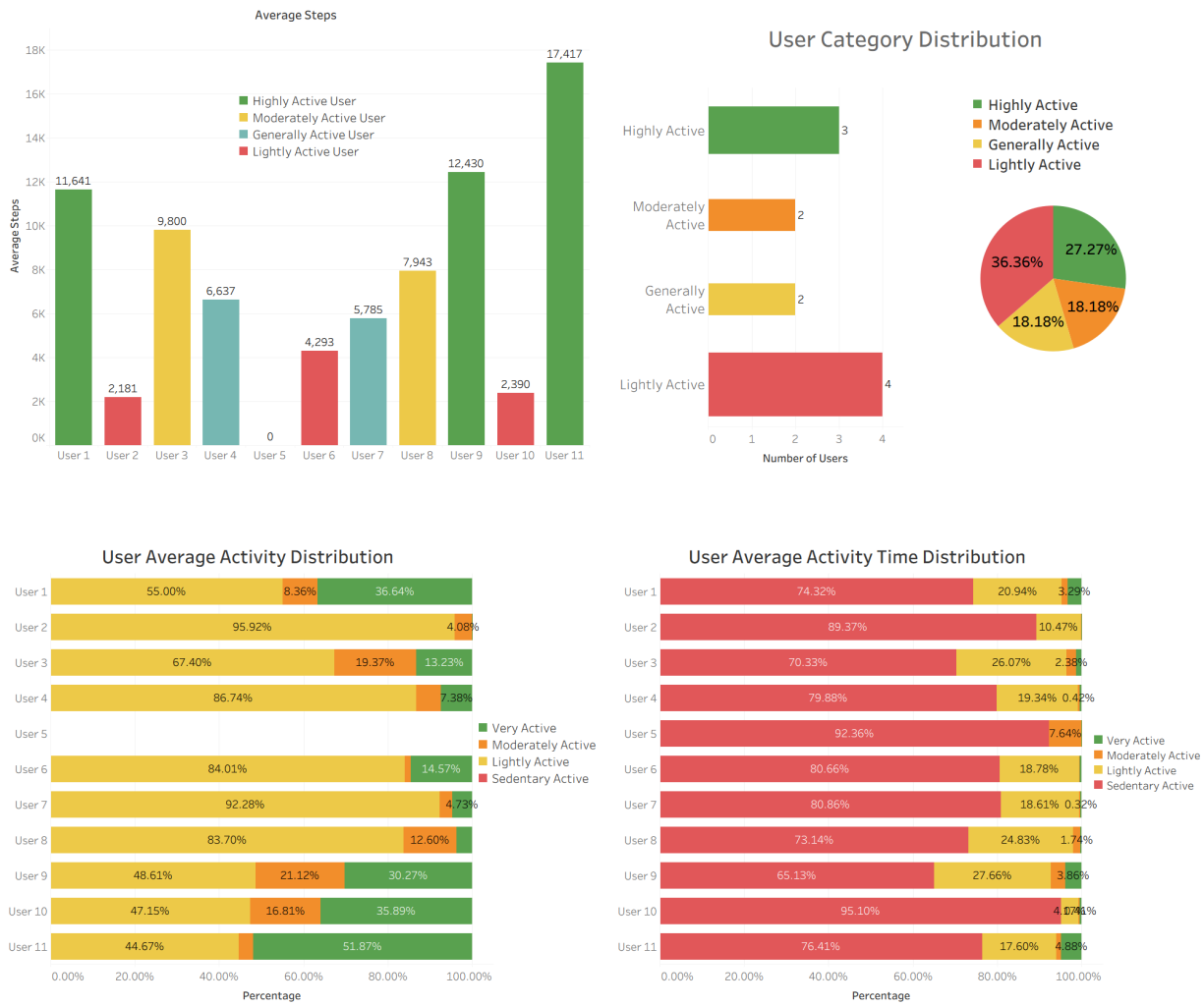

Figure 1: User Activity and Activity Time Summary

Based on average steps and exercise intensity data, the users are divided into 4 categories:

Table 1: User Activity Level Segmentation

Activity Level	Users	Average Step Range	Proportion	Feature
Highly Active	User 1, User 9, User 11	11,641-17,417	27.27%	High proportion of very active exercise
Moderately Active	User 3, User 8	7,943-9,800	18.18%	High proportion of moderately active exercise

Activity Level	Users	Average Step Range	Proportion	Feature
Generally Active	User 4, User 7	5,785-6,637	18.18%	High proportion of lightly active exercise
Lightly Active	User 2, User 5, User 6, User 10	0-4,293	36.36%	Almost no very active exercise

2. Key insights:

- **Highly active users are ideal users of the product**: User 11 is the top-performing user with an average of 17,417 daily steps and 51.87% very active share. User 1 and User 9 also have a very active share exceeding 30%. It's obvious that User 1, User 9, and User 11 are loyal users of the product.
- **Lightly active users represent the dominance of the user base**: Over 1/3 of users have an average fewer than 5,000 steps, with almost zero very active share (except User 11). These users are the core group to activate for future product development.
- **Positive correlation between activity level and exercise intensity**: Highly active users have a significantly higher share of very active exercise than lightly active users, while the opposite is true for the lightly active share. This finding demonstrates the rationality of the user segmentation.

Deep Analysis of User Exercise Behavior and Time Allocation

1. Exercise Intensity Distribution (User Average Activity Distribution)

- **Highly active users**: User 11 has a 51.87% very active share, while User 1 and User 9 have a very active share exceeding 30% and a relatively high moderately active share, with a lightly active share below 55%.
- **Lightly active users**: User 2, User 5, and User 6 have a lightly active share exceeding 83% and a very active share below 15%, while User 10 has a relative balance between lightly active share and very active share.
- **Moderately/Generally active users**: Both 2 types of users are dominated by lightly active exercise, while moderately active users (User 3 and User 8) have a larger moderately active share than generally active users (User 4 and User 7).

2. Exercise Time Allocation (User Average Activity Time Distribution)

- **Sedentary behavior dominates users' daily activity**: All users have a sedentary time share exceeding 65%. Among them, User 10 (95.10%), User 5 (92.36%), and User 2 (89.37%) have a sedentary share exceeding 85%, with almost no effective exercise time.
- **Reasonable time structure for highly active users**: Highly active users have a higher active time share (Very active + Moderately active + Lightly active) exceeding 20% and a relatively low sedentary share (User 9 has a share with only 65.13%).
- **Mismatch between exercise intensity and time**: Some users have high step counts but a large sedentary share, which indicates their exercise is concentrated in a short time period with high intensity, while daily life remains dominated by sitting. It's obvious that these users should optimize their activity habits.

3. Strong Correlation Between Steps and Calories Burned

The figures indicating the steps and calories burned are shown below by user category:

Figure 2: Walking Distance and Calories Consumption Summary (By User Category)

Based on the figures, it's evident that the trends of `Walking Steps` and `Calories Consumption` have a similar pattern for all users except user 5 (due to some potential mistakes during calories record). This validates the basic rationality of the data and reflects that exercise intensity is the core factor of calorie consumption.

However, it's unreasonable that User 3 and User 8 have similar walking steps, while User 8 burned more calories than User 3. Besides, User 5 burns more calories than User 10, while User 5 has exercise data with 0 steps. Thus, the company should try to improve the mechanism of calculating the calorie consumption based on different indicators.

Correlation Analysis Between Health Metrics (BMI/Weight) and Exercise Intensity

1. Correspondence Between Health Segmentation and Activity Level

After analyzing health data for all users, several useful figures are shown below based on the cleaned data:

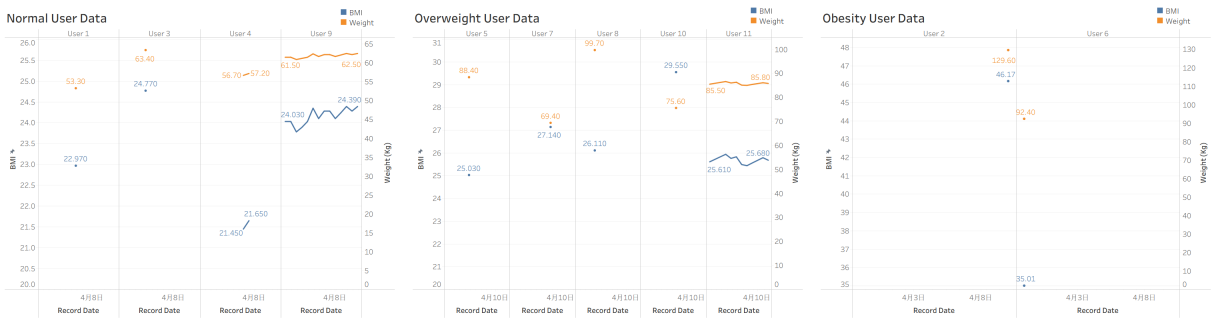

Figure 3: Health Data Summary

Based on health data for all users, the users are divided into 3 categories:

Table 2: User Health Level Segmentation and Corresponding Activity Level

Health Level	Users	BMI Range	Corresponding Activity Level	Feature
Normal	User 1, User 3, User 4, User 9	21.45-24.77	Highly/Moderately Active	Stable weight with low BMI
Overweight	User 5, User 7, User 8, User 10, User 11	25.03-29.55	Moderately/Generally Active	Slight weight excess with slightly high BMI
Obese	User 2, User 6	35.01-46.17	Generally/Lightly Active	Severe weight excess with high BMI

2. Core Correlation Insights:

- **Strong negative correlation between BMI and activity level**: The higher the user's exercise intensity and very active share, the lower their BMI and the better their health status.
- **Lightly active users needs more health intervention to keep healthy**: Both obese users are in the lightly active level, and insufficient exercise is the core cause of weight excess, which requires a comprehensive exercise plan to inspire obese users to be healthy.
- **Better health status for highly active users**: Among the three highly active users, two of them (User 1, User 9) are of normal weight, and only one (User 11) is overweight with a better health performance and proportion than other user levels.

Data Quality Analysis

1. Abnormal Data Identification

When analyzing the walking distance data, several users have data deviations that seem inaccurate when collecting the data:

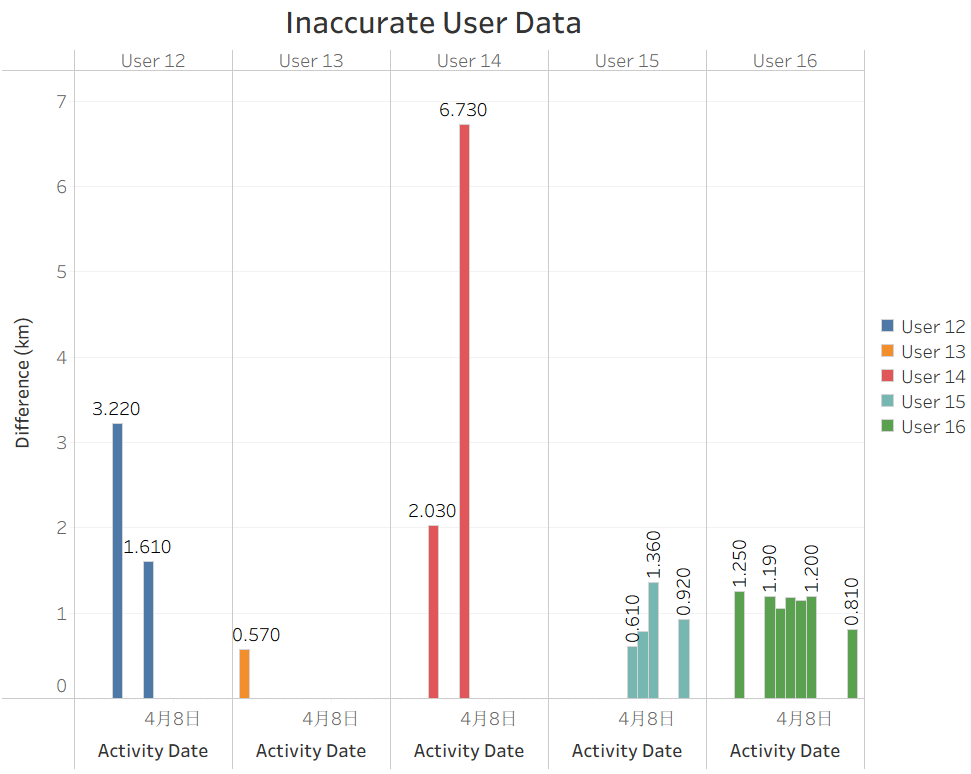

Figure 4: Inaccurate User Data

Based on the figure above, it's obvious that:

- **User 14 has the largest deviation**: User 14 has a significant data difference of 6.730 km, which is a severely abnormal deviation. It is possibly caused by device failure, manual mistake, or incorrect wearing.
- **User 12 has the second largest deviation**: User 12 has a difference of 3.220 km, which is about half of User 14's deviation. The reason might also be the same as User 14.
- Other users (User 13, 15, and 16) have deviations within 2 km, which are mild deviation values.

2. Product Optimization Directions

- **Data validation mechanism**: Optimize data correction algorithms to reduce extreme outliers and improve data accuracy.
- **Anomaly alert function**: Add data anomaly alerts for users to help avoid device-wearing issues and enhance user experience.
- **Data cleaning mechanism**: Establish an anomaly data filtering mechanism in the back-end to improve report reliability and precision.

Product Use and Operation Optimization Recommendations

1. Precision Operation Suggestions for Different User Levels

- **Highly active users**: Build "exercise benchmarks", launch challenge tasks and badge systems to strengthen user stickiness; explore their exercise habits to provide references for other users.
- **Moderately/Generally active users**: Set suitable goals to guide users to increase their very active exercise share and reduce sedentary time. Organize personalized exercise plans to gradually improve activity levels.
- **Lightly active users**: Set check-in incentives and sedentary reminder functions to improve sedentary issues. Customize low-intensity entry exercise plans for obese/overweight users to activate their training passion.

2. Product Function Optimization

- **Sedentary reminder function**: Strengthen sedentary reminders and push fragmented exercise suggestions to help users reduce daily sedentary time.
- **Health data report**: Summarize exercise data and BMI/weight data to generate personalized health reports to provide a comprehensive exercise-health service for users.
- **Data quality improvement**: Optimize device precision, add anomaly data validation function, and enhance data credibility.

3. Business Growth Directions

- **Lightly active user activation**: Over 1/3 of lightly active users are the core potential for product growth. Activating them through personalized operations can significantly improve overall activity and user satisfaction.
- **Health data connection**: Based on the strong correlation between exercise and health, it's beneficial to constitute a core selling point of "exercise health management" to enhance product competitiveness in the market.
- **User Segmentation services**: Achieve precision push based on user activity level and health indicators to improve personalized product experience.

Final Summary

This analysis comprehensively illustrates the exercise behavior characteristics of users:

- Highly active users have good exercise habits and excellent health status, which makes them the core users of the product.
- Lightly active users have insufficient exercise, severe sedentary behavior, and high health risks, which makes them the key activation target of the product.

Meanwhile, the outliers in the data also reflect room for optimization in the product's data validation.

Through personalized user operations and product function optimization, we can effectively increase the product usage frequency of lightly active users, improve user health status, and achieve significant enhancement of product value and user health.